Date:29-1-26

DS P9:Perform one-way ANOVA to compare means across multiple groups. Conduct post-hoc test to identify significant differences between group means.

In [2]:
```
import pandas as pd
import scipy.stats as stats
from statsmodels.stats.multicomp import pairwise_tukeyhsd
```

In [3]:
```
grup1=[23,25,29,34,30]
grup2=[19,20,22,24,25]
grup3=[15,18,20,21,17]
grup4=[28,24,26,30,29]

data=pd.DataFrame({'value':grup1+grup2+grup3+grup4,
                   'group':['Group1']*len(grup1)+['Group2']*len(grup2)+['Group3']*len(grup3)+['Group4']*len(grup4)})
```

In [4]:
```
#perform one way anova
f_statistics,p_value=stats.f_oneway(grup1,grup2,grup3,grup4)
print("One way ANOVA:")
print("F statistics: ",f_statistics)
print("P value: ",p_value)
```

```
One way ANOVA:
F statistics:  12.139872842870115
P value:  0.00021465200901629603
```

In [5]:
```
#perform tukey-kramer post hoc test
tukey_results=pairwise_tukeyhsd(data['value'],data['group'])
print("\nTukey-Kramer Post-hoc test")
print(tukey_results)
```

```
Tukey-Kramer Post-hoc test
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1 group2 meandiff p-adj   lower    upper  reject
-----------------------------------------------------
Group1 Group2     -6.2  0.024 -11.6809 -0.7191   True
Group1 Group3    -10.0 0.0004 -15.4809 -4.5191   True
Group1 Group4     -0.8 0.9747  -6.2809  4.6809  False
Group2 Group3     -3.8 0.2348  -9.2809  1.6809  False
Group2 Group4      5.4 0.0542  -0.0809 10.8809  False
Group3 Group4      9.2  0.001   3.7191 14.6809   True
-----------------------------------------------------
```